

Nathan Potraz

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CORE COMPETENCIES

- Go & REST API Development
- React & React Native
- Systems Programming
- Docker & CI/CD
- SQL & Database Design
- Hardware Engineering & QA
- TypeScript / JavaScript
- Technical Communication

EXPERIENCE

CadraWorks June 2022 - Present

Hardware & Software Engineer

Oceanside, CA

- Built and operated a custom controller business - **100+ units shipped at \$150-\$400/unit** with a one-week order-to-ship turnaround.
- Engineered performance-critical hardware for professional esports clients: circuit-board soldering, firmware optimization, and millisecond-sensitive tuning with **zero tolerance for client-facing failures**.
- First tester on an open-source controller motherboard project - validated board connections, reviewed documentation, and produced a **calibration guide with 10,000+ views**.

PROJECTS

Devlog Go | TypeScript | React | Docker | Cloud Run | CI/CD github.com/ntpotraz/devlog 2025

Full-stack journaling platform with immutable post versioning. Go HTTP server with type-safe SQL (sqlc), Goose migrations, Clerk JWT middleware, and a React/Vite SPA served from the same binary; Docker + GitHub Actions CI/CD, deployed to Google Cloud Run.

PIC Health App React Native | Expo | Google Calendar API github.com/ntpotraz/PIC-Health-App 2024

Led a 5-person team through a university capstone with a health research org - ran weekly sponsor meetings, owned the roadmap, and demoed each milestone to non-technical stakeholders. Built the full stack: React Native + Expo cross-platform pipeline targeting iOS, Android, and Web from a single codebase.

Japanese/English Translator C++ | DFA Scanner | Recursive Parser github.com/ntpotraz/cs421-translator 2024

Three-phase language translator in C++ - DFA-based scanner (derived from NFA diagrams), recursive descent parser, and a translation layer backed by a lexicon map; demonstrates formal language theory and compiler internals end-to-end.

Hack Computer Toolchain Java | Compiler | VM Translator | Assembler 2020

Two-stage compilation pipeline for the Hack computer architecture: a VM Translator (stack bytecode to assembly) and an Assembler (symbolic assembly to 16-bit binary), with full VM instruction set support and two-pass symbol resolution.

DNS Resolver Python | UDP Sockets | Networking github.com/Saltair1/NetworkPythonProject 2023

Three-tier DNS resolution hierarchy over raw UDP sockets with hand-constructed binary DNS packets. Implements recursive query forwarding, transaction ID validation, and TTL-based cache invalidation across A, AAAA, CNAME, and NS record types.

Gloomy Monsters React | Vite | JavaScript ntpotraz.github.io/gloomy-monsters 2022

Live Gloomhaven companion app - search, add, and track every monster on the board with elite/non-elite toggling and per-unit HP tracking. Component-tree architecture (Board > MonsterPack > Monster) driven by a bundled JSON dataset, deployed via Vite and gh-pages automation.

EDUCATION

B.S. Computer Science - California State University, San Marcos Dec 2024

Cum Laude | 3.7 GPA

TECHNICAL SKILLS

Languages: Go, TypeScript, C++, Java, Python, PHP, SQL Frontend: React, Vite, Expo, React Native, JavaScript, CSS3

Backend/Systems: Node.js, Bun, Docker, raw UDP sockets, binary protocol construction

Tools: Git, GitHub Actions, Biome, Maven, sqlc, Goose Hardware: Circuit-board soldering, firmware tuning, hardware validation, calibration